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CypherCon 2026

WarGames for Defenders

Playing Threats with MITRE ATT&CK Navigator

March 31, 2026

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Thomas Freeman

CISSP, C|CISO, GPEN, GCIA, GCIH, GCWN

DIRECTOR

- Leads CISO Advisory services
- 25+ years of IT and cybersecurity experience
- 20+ years in higher education

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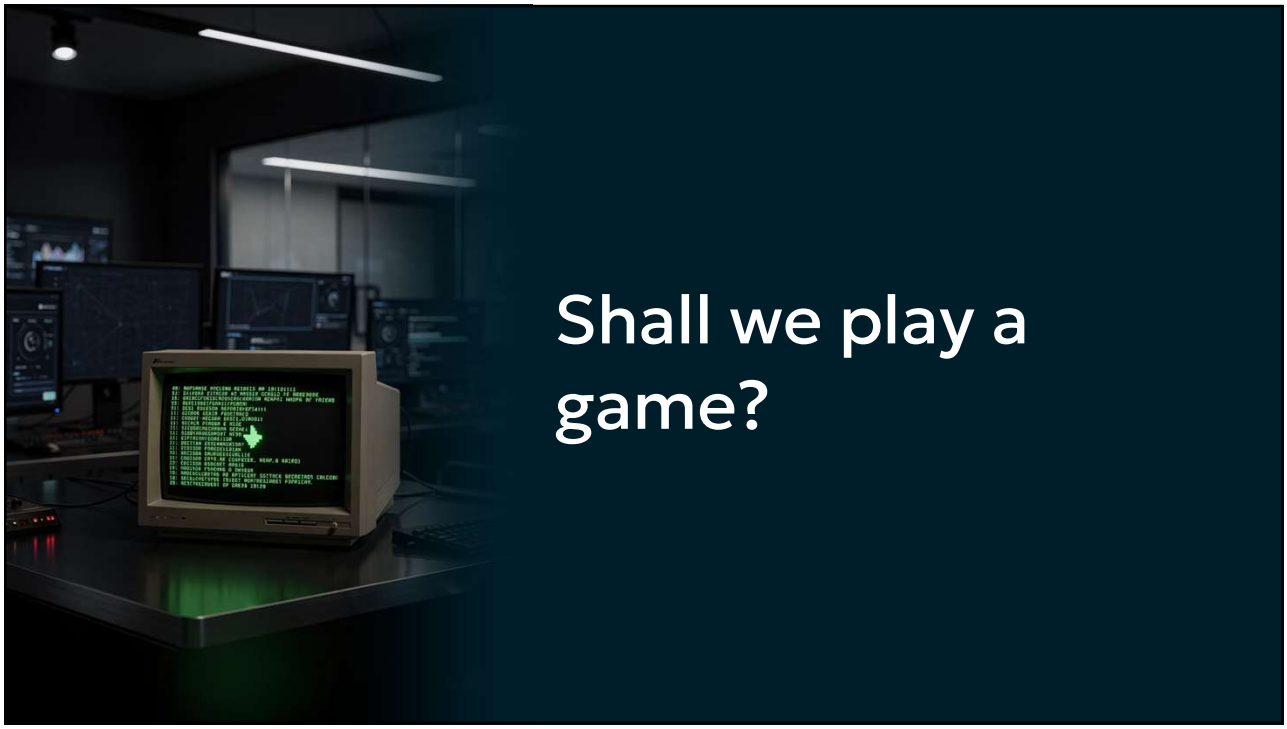
Shall we play a game?

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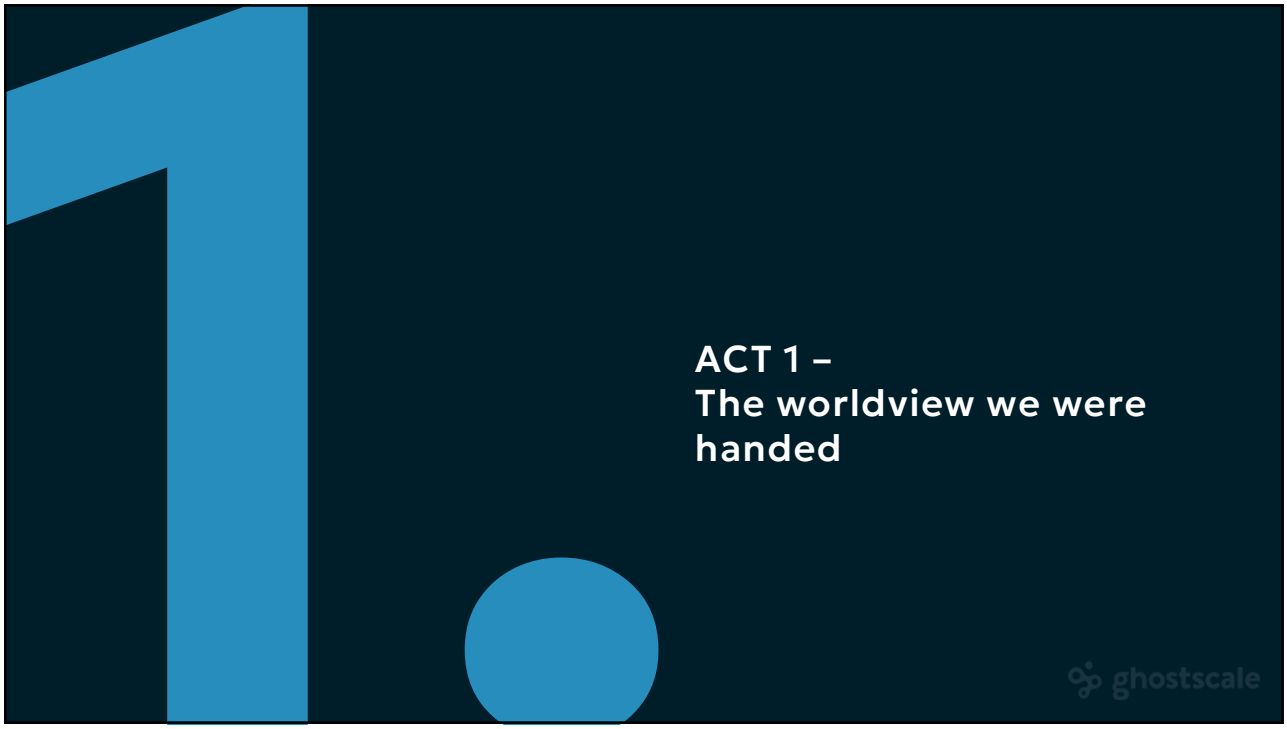
1.
ACT 1 –
The worldview we were handed
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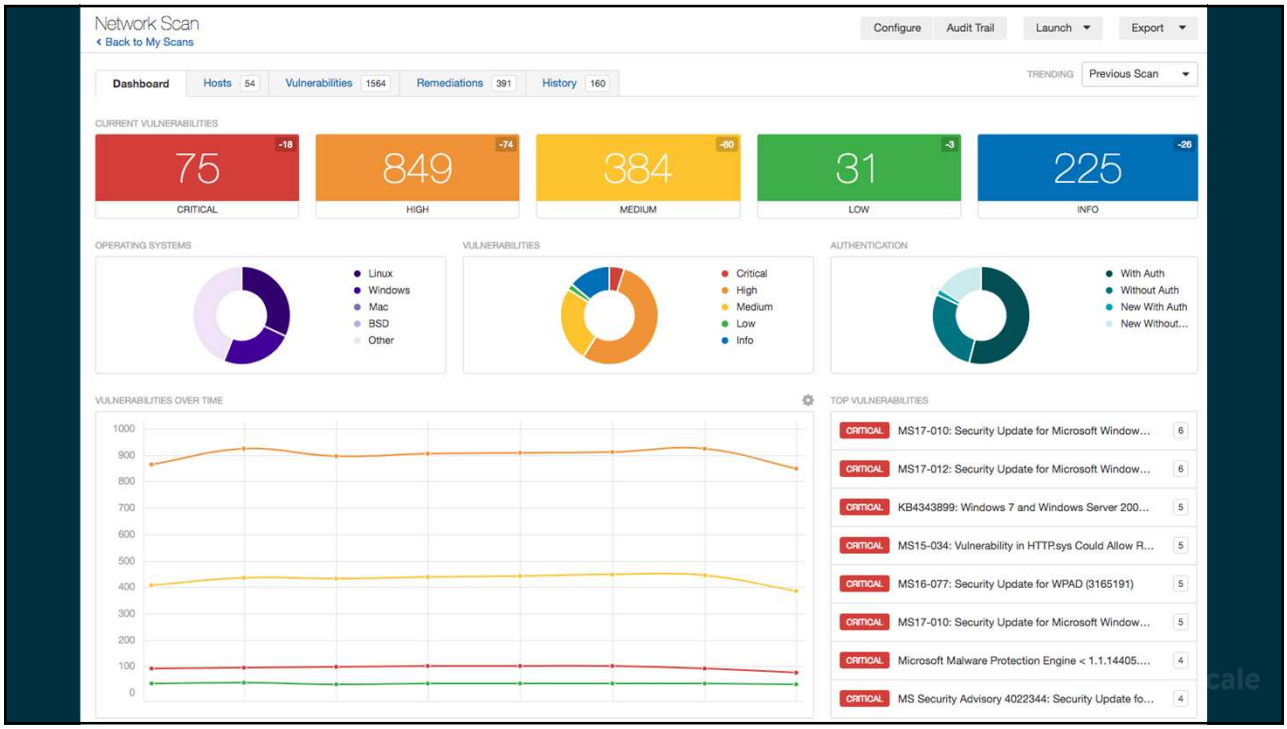
Network Scan
Back to My Scans
Configure
Audit Trail
Launch
Export
Dashboard
Hosts 54
Vulnerabilities 1564
Remediations 391
History 160
TRENDING
Previous Scan
CURRENT VULNERABILITIES
75 -18 CRITICAL
849 -74 HIGH
384 -60 MEDIUM
31 -3 LOW
225 -26 INFO
OPERATING SYSTEMS
Linux
Windows
Mac
BSD
Other
VULNERABILITIES
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Info
AUTHENTICATION
With Auth
Without Auth
New With Auth
New Without...
VULNERABILITIES OVER TIME
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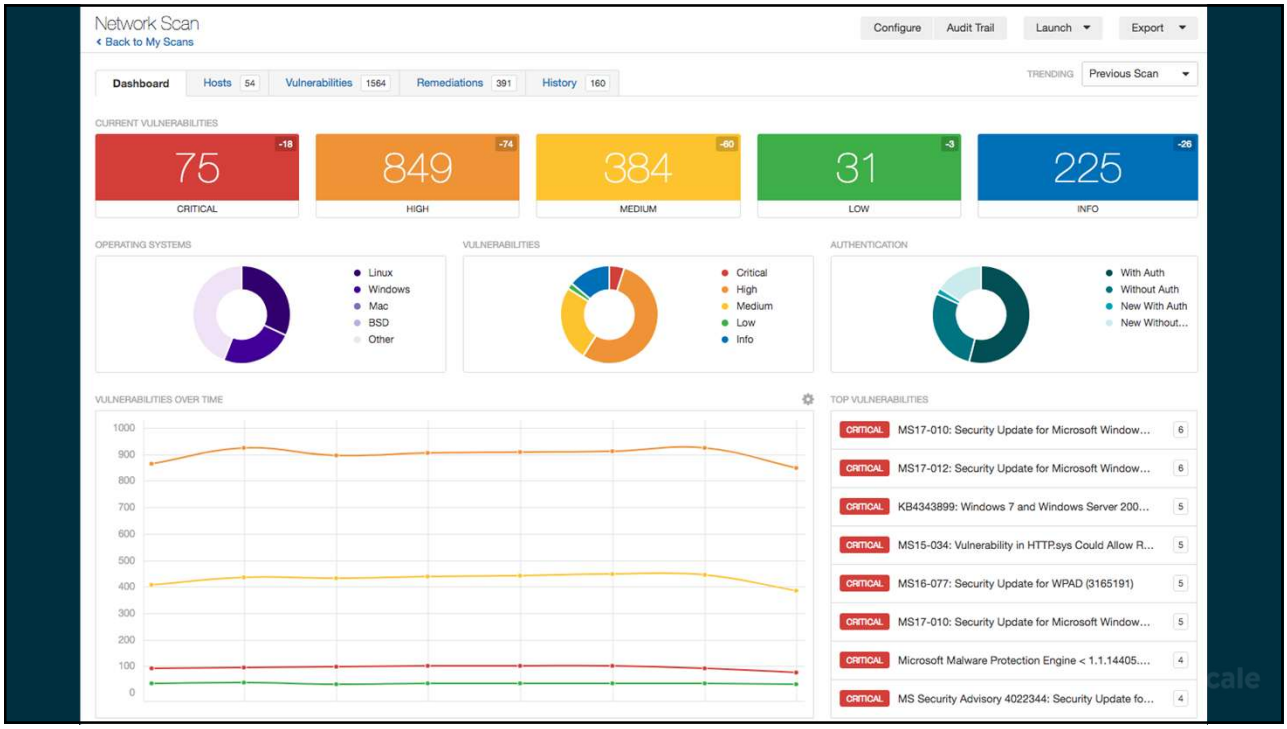
TOP VULNERABILITIES
CRITICAL MS17-010: Security Update for Microsoft Window... 6
CRITICAL MS17-012: Security Update for Microsoft Window... 6
CRITICAL KB4343899: Windows 7 and Windows Server 200... 5
CRITICAL MS15-034: Vulnerability in HTTP.sys Could Allow R... 5
CRITICAL MS16-077: Security Update for WPAD (3165191) 5
CRITICAL MS17-010: Security Update for Microsoft Window... 5
CRITICAL Microsoft Malware Protection Engine < 1.1.14405.... 4
CRITICAL MS Security Advisory 4022344: Security Update fo... 4

5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Severity Is Not Risk

CVSS Score

Measures vulnerability severity (cold, mechanical, numbers)

Adversary Behavior

Measures what is **actually coming for you.** (human, intentional, motivated.)

- These have never been the same thing.

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There is a better way

> "A strange game. The only winning move is not to play."

Joshua ran every scenario. Every strategy. Every counter.
The conclusion: don't play the adversary's game; **know it before they play it.**

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2.

ACT 2 –
Changing the view

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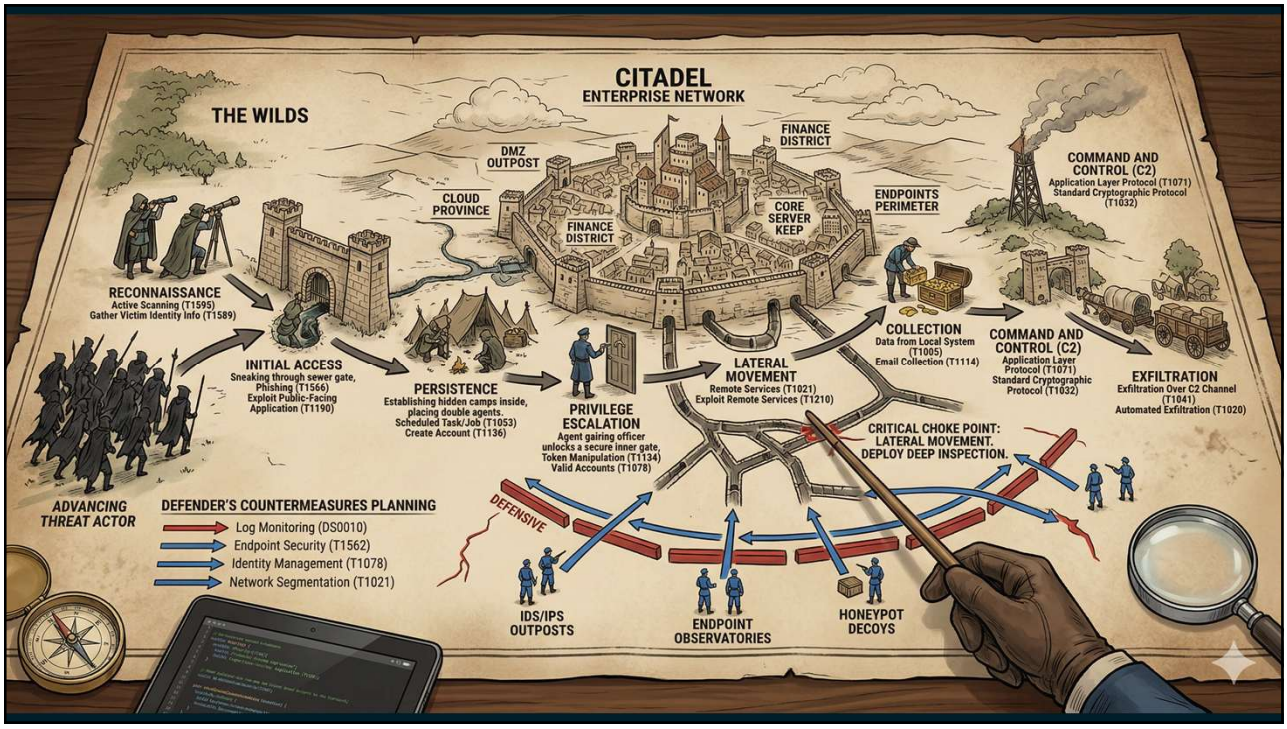

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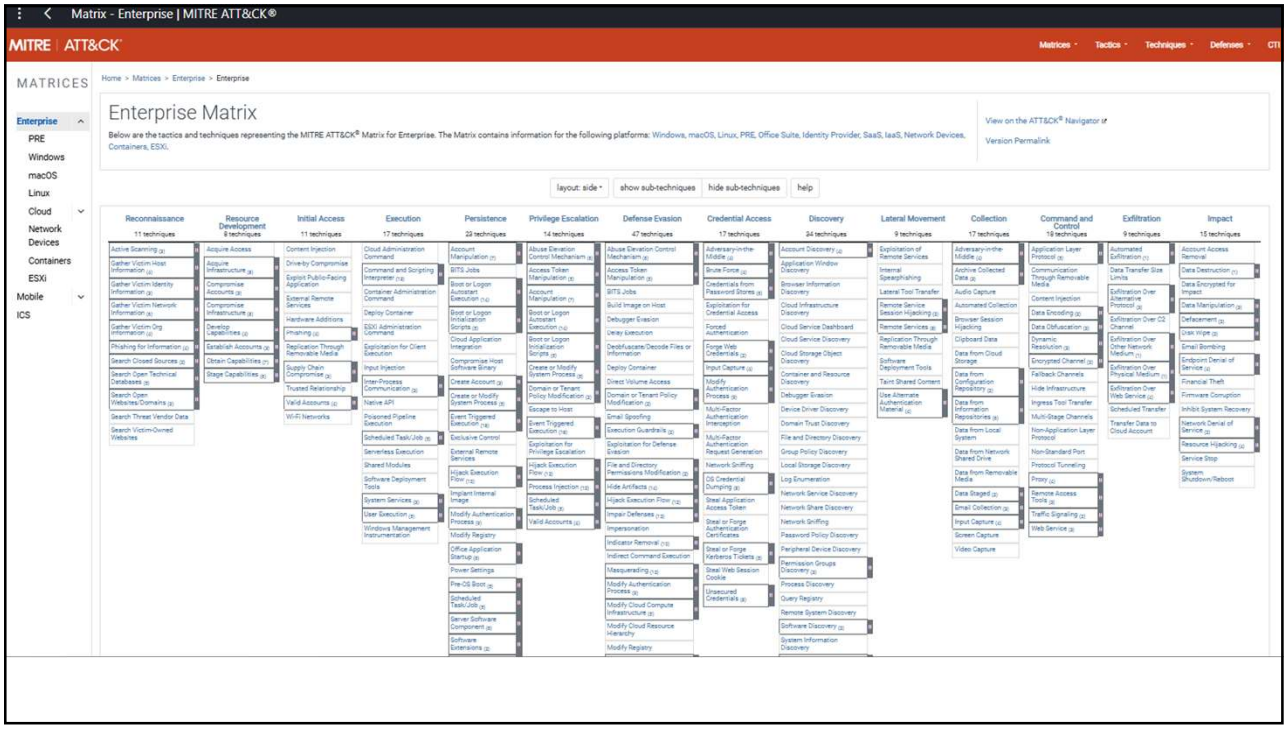
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ESXi
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ICS
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Enterprise Matrix
View on the ATT&CK® Navigator
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help
Reconnaissance
Resource Development
Initial Access
Execution
Persistence
Privilege Escalation
Defense Evasion
Credential Access
Discovery
Lateral Movement
Collection
Command and Control
Exfiltration
Impact

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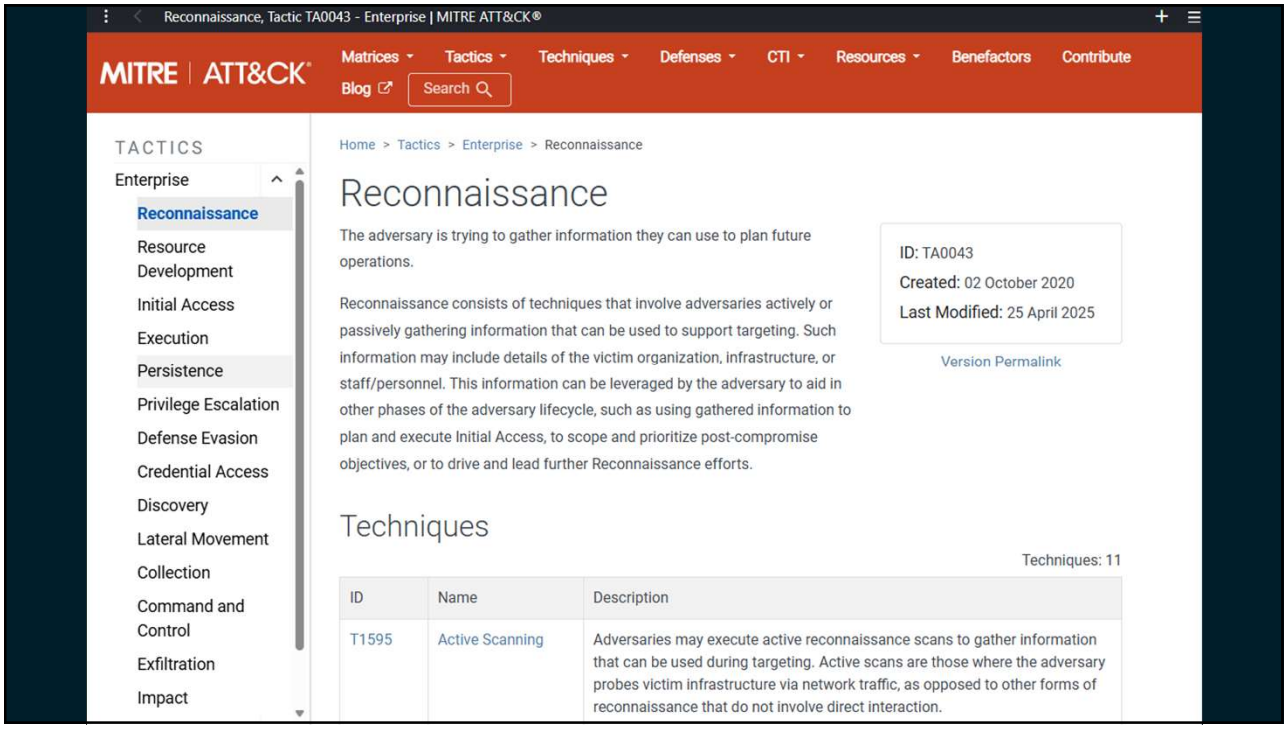
Reconnaissance, Tactic TA0043 - Enterprise | MITRE ATT&CK®
MITRE | ATT&CK
Matrices
Tactics
Techniques
Defenses
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Resources
Benefactors
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Search
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Enterprise
Reconnaissance
Resource Development
Initial Access
Execution
Persistence
Privilege Escalation
Defense Evasion
Credential Access
Discovery
Lateral Movement
Collection
Command and Control
Exfiltration
Impact
Home > Tactics > Enterprise > Reconnaissance
Reconnaissance
The adversary is trying to gather information they can use to plan future operations.
Reconnaissance consists of techniques that involve adversaries actively or passively gathering information that can be used to support targeting. Such information may include details of the victim organization, infrastructure, or staff/personnel. This information can be leveraged by the adversary to aid in other phases of the adversary lifecycle, such as using gathered information to plan and execute Initial Access, to scope and prioritize post-compromise objectives, or to drive and lead further Reconnaissance efforts.
ID: TA0043
Created: 02 October 2020
Last Modified: 25 April 2025
Version Permalink
Techniques
Techniques: 11
ID
Name
Description
T1595
Active Scanning
Adversaries may execute active reconnaissance scans to gather information that can be used during targeting. Active scans are those where the adversary probes victim infrastructure via network traffic, as opposed to other forms of reconnaissance that do not involve direct interaction.

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Intelligence Before Engagement

They built the adversary profile first. **Then** designed the operation. We have the tools to do exactly that.

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3.

Threat Actor Persona

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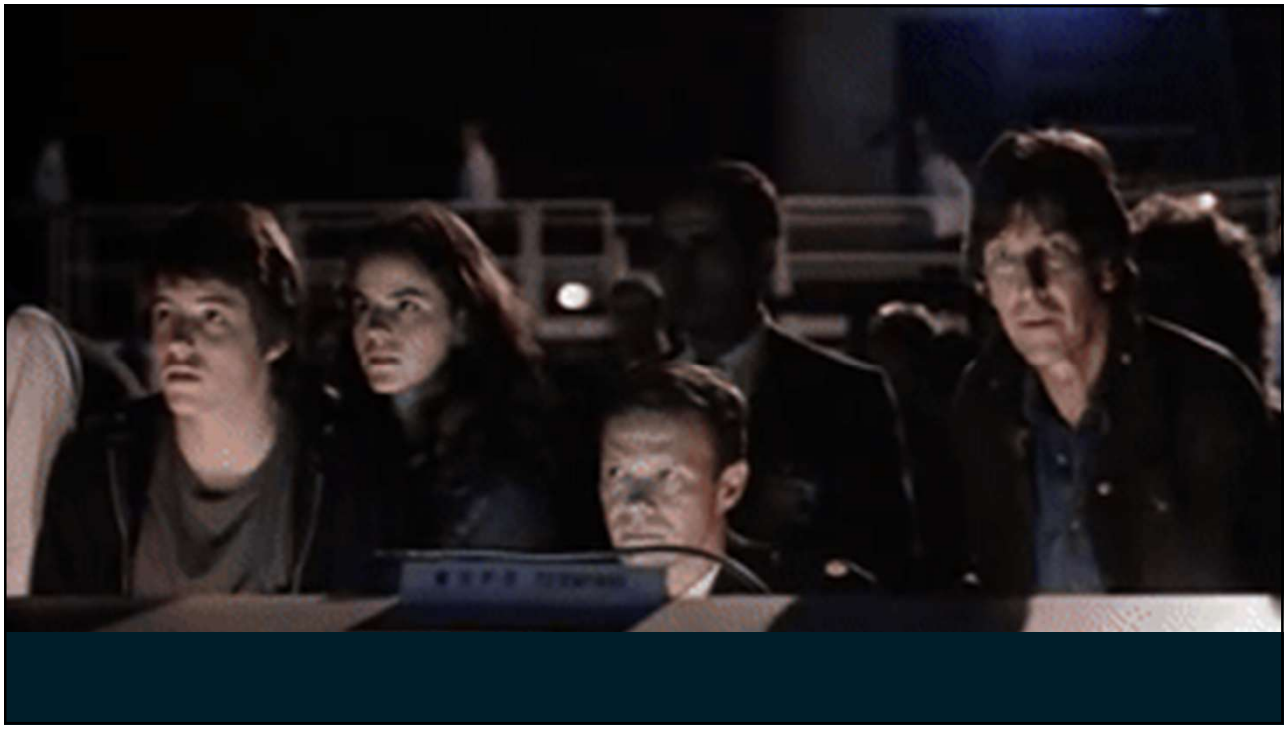

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Joshua is our scouting report on the enemy, built from real attacks on companies like ours, so every security decision we make is aimed at the threat that is actually coming for us

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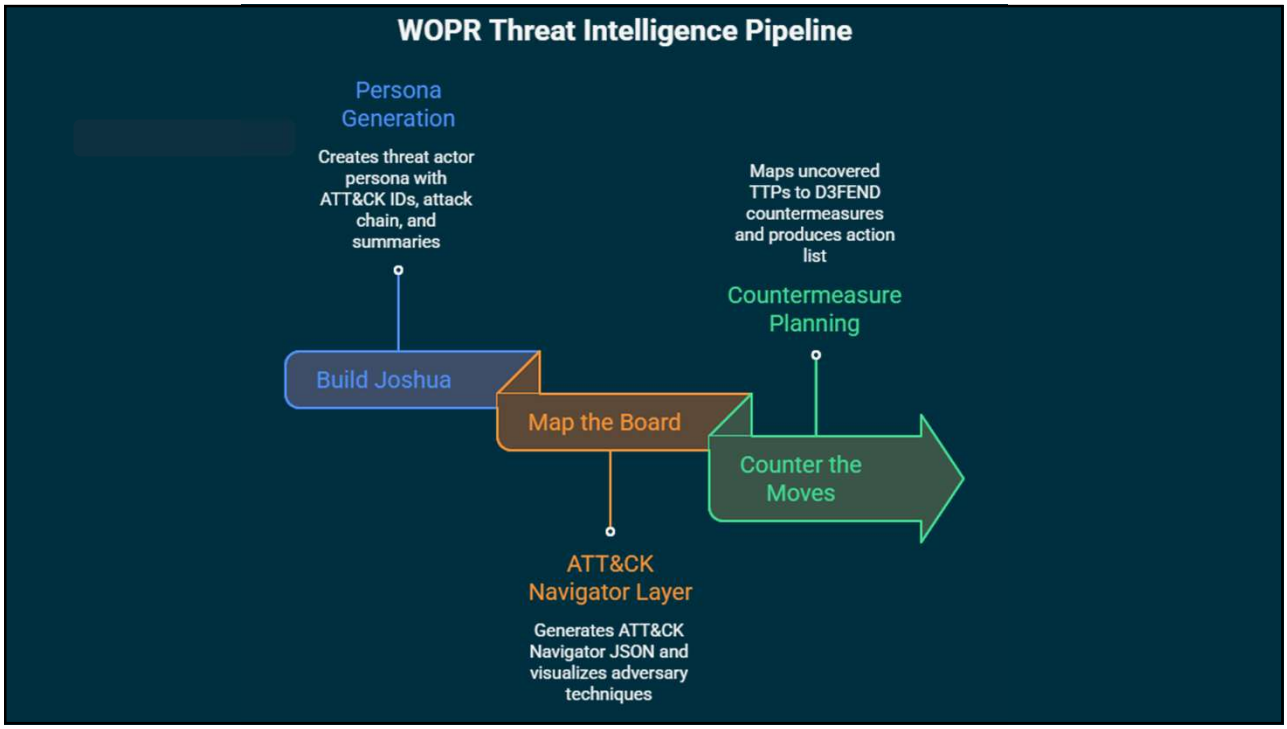

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Falken Holdings

Mid-market. Three entities. One threat actor watching.

Falken Holdings
Parent Company · 520 employees · Hybrid infrastructure

Falken Property Group
Real estate development

Ridgeline Manufacturing
Never fully integrated

Summit Professional Services
Client PII & M&A data

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Building Joshua

This is your **scouting report.**

Threat Type

Ransomware Affiliate / Initial Access Broker

Target Industries

Manufacturing · Professional Services · Financial Holdings

Primary TTPs

Phishing · Valid Accounts · Credential Dumping · Lateral Movement

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Joshua On the Board

Reconnaissance	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Exfiltration	Impact
	Phishing T1566					OS Credential Dumping T1003		Remote Services T1021			Data Encrypted for Impact T1486
	Valid Accounts T1078					Brute Force T1110		Pass the Hash T1550			Inhibit System Recovery T1490
	External Remote Services T1133										

Highlighted red cells show threat actor Joshua's confirmed TTPs, concentrated in Initial Access, Credential Access, Lateral Movement, and Impact tactics.

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The Gap Analysis

GREEN — Covered
Control in place. Joshua stopped here.

YELLOW — Partial
Control exists. Insufficient. Needs work.

RED — Gap
No control. Joshua wins here. Right now. Today.

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D3FEND — The Counter Measures

ATT&CK — ATTACK
T1566 Phishing → Initial Access vector confirmed for Joshua

D3FEND — DEFENSE
Message Analysis / User Training → Mapped to your control framework

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Joshua Drives Everything

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Map Joshua against your whole program

TTP	Risk Reg.	Controls	Policies	IR Plan	Tabletop	Vuln Mgmt	Training	Board
T1566 Phishing	Partial	Yes	Yes	Partial	Gap	Partial	Yes	Gap
T1078 Valid Accounts	Gap	Partial	Partial	Gap	Gap	Partial	Partial	Gap
T1133 Ext. Remote Svc	Partial	Yes	Gap	Gap	Gap	Yes	Gap	Gap
T1003 Cred. Dumping	Gap	Partial	Gap	Gap	Gap	Partial	Gap	Gap
T1021 Lateral Movement	Gap	Partial	Gap	Partial	Gap	Gap	Gap	Gap
T1486 Ransomwar e	Yes	Partial	Yes	Partial	Partial	Gap	Partial	Partial
T1490 Shadow Copy Del.	Gap	Gap	Gap	Gap	Gap	Gap	Gap	Gap

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4.

Demo
https://github.com/zoeified/wopr

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Monday, when you are back in the office.

- CREATE YOUR AI PROJECT
- FILL OUT YOUR AI CONTEXT
- CREATE YOUR JOSHUA
- IMPORT JOSHUA INTO NAVIGATOR
- CREATE ASSESSMENT LAYERS
- PLAN YOUR D3FENSE

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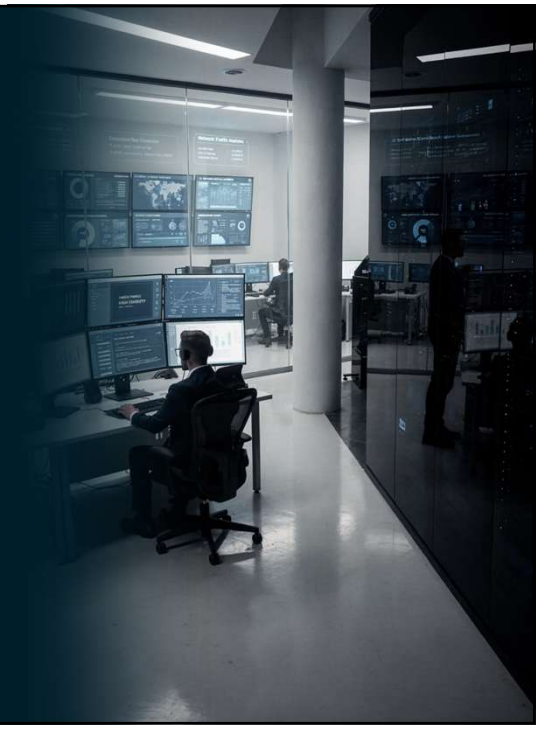
Build your Joshua before Joshua builds you.



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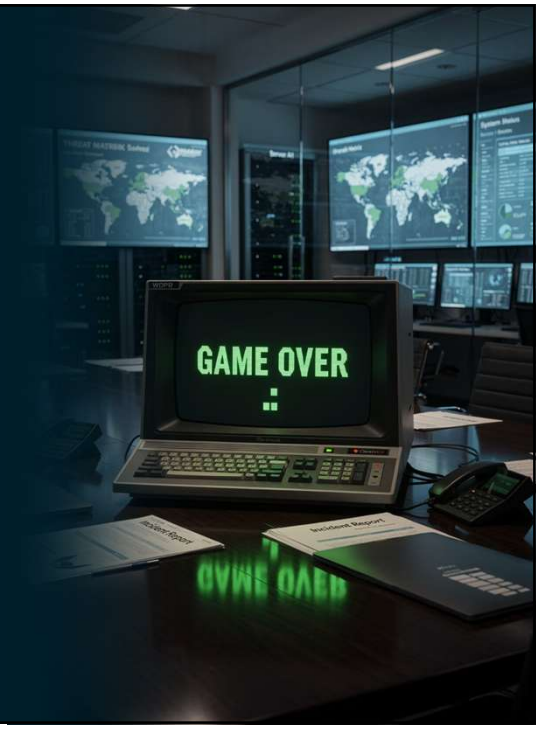
"A strange game…

The only winning move is…"



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TODAY'S PRESENTER

Thomas Freeman, Director
CISSP, C|CISO, GPEN, GCIA, GCIH, GCWN
Thomas.Freeman@ghostscale.com
262.527.4746

https://github.com/zoeified/wopr

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